Enhancing Pre-Training Data Detection through Distribution Shape Analysis: A Multi-Scale Weighted Residual Approach to Min-K%++

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Abstract

Pre-training data detection in large language models has emerged as a critical challenge for model transparency and compliance, with membership inference attacks serving as the primary mechanism for identifying whether specific text sequences were part of a model's training data. While Min-K%++ represents the current state-of-the-art approach, it suffers from a fundamental limitation: uniform aggregation of token-level scores ignores valuable distributional patterns that could enhance detection accuracy. We propose a novel enhancement through residual score decomposition with multi-scale importance weighting, which analyzes distribution shape features such as skewness, kurtosis, and entropy to reveal training versus non-training patterns. Our method decomposes Min-K%++ scores into trend and residual components using exponential moving averages (Lucas & Saccucci, 1990), applies position-based weighting that emphasizes earlier tokens in sequences, and performs multi-scale deviation analysis to capture patterns across different temporal scales. Extensive experiments on WikiMIA (Shi et al., 2024) across multiple sequence lengths (32, 64, 128 tokens) and model architectures (Pythia-2.8b (Biderman et al., 2023), Mamba-1.4b (Gu & Dao, 2023)) demonstrate consistent improvements up to 1.6 percentage points AUROC, with the largest gains observed for longer sequences where positional patterns become more distinctive. Our approach requires minimal computational overhead and provides interpretable insights into how distributional properties correlate with membership detection quality.

1 Introduction

Large language models raise concerns about data transparency and intellectual property compliance (Achiam et al., 2023; Touvron et al., 2023), motivating membership inference attacks (MIAs) (Shokri et al., 2017; Carlini et al., 2022a) to determine whether specific text sequences were in training data. Recent advances have moved beyond confidence-based metrics (Carlini et al., 2021; Watson et al., 2022) toward likelihood-based approaches (Mireshghallah et al., 2022; Mattern et al., 2023; Xie et al., 2024). Min-K%++ (Zhang et al., 2025) represents the current state-of-the-art, grounding its approach in score matching theory (Hyvärinen & Dayan, 2005; Koehler et al., 2022). However, Min-K%++ suffers from uniform aggregation of token-level scores that ignores valuable distributional patterns.

Our key insight is that distribution shape features contain valuable membership signals overlooked by uniform aggregation (Gehrmann et al., 2019; Liu et al., 2020). Training data typically shows more concentrated patterns while non-training data displays heavier tails (Carlini et al., 2018, 2022b). Position-dependent weighting makes intuitive sense as early tokens establish domain and style context that models strongly associate with training patterns.

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We propose enhancing Min-K%++ through residual score decomposition with multi-scale importance weighting. Our approach includes: (1) exponential moving average trend analysis decomposing scores into trend and residual components, (2) position-based weighting recognizing varying token informativeness, and (3) multi-scale deviation analysis capturing patterns across temporal scales. These enhancements require minimal computational overhead as they operate on pre-computed Min-K%++ scores.

We evaluate on WikiMIA (Shi et al., 2024) across sequence lengths (32, 64, 128 tokens) and architectures (Pythia-2.8b (Biderman et al., 2023), Mamba-1.4b (Gu & Dao, 2023)). Results show consistent improvements, with linear decay position weighting achieving up to 1.6 percentage point AUROC gains. Ablation studies reveal position-based weighting as the primary driver.

Our contributions include: (1) identifying distribution shape analysis as fundamental for improving membership inference with theoretical motivation and empirical validation, (2) developing a practical method enhancing Min-K%++ through residual decomposition and adaptive weighting while maintaining efficiency, and (3) extensive experiments demonstrating robustness across models and sequence lengths with detailed ablation studies.

2 Related Work

Membership Inference Foundations. Membership inference attacks (Shokri et al., 2017) exploit information leakage to identify training data. Early confidence-based approaches (Carlini et al., 2021; Watson et al., 2022) proved inadequate, motivating reference-aware methods (Mireshghallah et al., 2022; Mattern et al., 2023; Fu et al., 2023).

The paper fails to discuss the differences between the problem settings of common membership inference attacks and those addressed by Min-k% or Min-k%++, which should be an important part.

Min-K%++ and Core Limitation. Min-K%++ (Zhang et al., 2025) achieves robust performance through score matching theory by aggregating the k% lowest-scoring tokens. However, it suffers from *uniform aggregation*, treating all selected tokens equally and ignoring valuable distributional patterns.

Distributional Analysis. Prior work demonstrates distributional patterns' value: Gehrmann et al. (2019) showed token-rank histograms reveal machine-generated text patterns, Liu et al. (2020) demonstrated energy-based scores outperform confidence approaches, and statistical process control (Lucas & Saccucci, 1990) uses exponentially weighted moving averages for shift detection. Recent methods like ReCaLL (Xie et al., 2024) and self-prompt calibration (Fu et al., 2023) rely on scalar aggregation, ignoring distributional patterns despite sequence positions carrying varying information content (Vaswani et al., 2017).

Our Contribution and Differentiation. Our work addresses the core limitation of uniform aggregation in Min-K%++ by introducing *residual score decomposition with multi-scale importance weighting*. Unlike methods that develop entirely new scoring schemes, we enhance the proven Min-K%++ foundation by: (1) decomposing scores into trend and residual components to identify tokens that deviate from local patterns, (2) applying position-based weighting that recognizes varying token informativeness, and (3) performing multi-scale analysis to capture patterns across different temporal scales. This approach directly targets the distributional blind spots of uniform aggregation while maintaining the theoretical grounding and computational efficiency that make Min-K%++ effective.

Citation: There is no mention about Min-K% [Shi+, ICLR2024].

3 Method

3.1 Overview

We first introduce the baseline Min-K%++ method, then present our enhancement through residual score decomposition with multi-scale token importance weighting.

3.2 Preview of Baseline Method

Min-K%++ (Zhang et al., 2025) represents the current state-of-the-art in membership inference attacks for large language models. The method is grounded in score matching theory and provides a theoretically motivated approach to pre-training data detection.

3.2.1 Theoretical Foundation

The core insight of Min-K%++ stems from the relationship between maximum likelihood training and implicit score matching (Lin et al., 2015; Kim et al., 2022). For continuous distributions, the maximum likelihood objective can be reformulated using implicit score matching (ISM) as:

$$\frac{1}{N}\sum_{x}\left[\frac{1}{2}||\psi(x)||^2 + \sum_{i=1}^{d}\frac{\partial\psi_i(x)}{\partial x_i}\right], \tag{1}$$

where $\psi(x) = \frac{\partial \log p(x)}{\partial x}$ is the score function. This formulation reveals that maximum likelihood training implicitly minimizes both the magnitude of first-order derivatives and the sum of second-order partial derivatives of $\log p(x)$. Consequently, training samples tend to form local maxima or locate near local maxima along each input dimension.

3.2.2 Method Formulation

Translating this insight to the discrete categorical distribution of LLMs, Min-K%++ computes a normalized score for each token position:

$$\text{Min-K\%++}_{\text{token}}(x_{<t}, x_t) = \frac{\log p(x_t|x_{<t}) - \mu_{\cdot|x_{<t}}}{\sigma_{\cdot|x_{<t}}}, \tag{2}$$

$$\text{Min-K\%++}(x) = \frac{1}{|\text{min-}k\%|}\sum_{(x_{<t},x_t)\in\text{min-}k\%}\text{Min-K\%++}_{\text{token}}(x_{<t}, x_t). \tag{3}$$

Here, $\mu_{\cdot|x_{<t}} = \mathbb{E}_{z\sim p(\cdot|x_{<t})}[\log p(z|x_{<t})]$ and $\sigma_{\cdot|x_{<t}} = \sqrt{\mathbb{E}_{z\sim p(\cdot|x_{<t})}[(\log p(z|x_{<t}) - \mu_{\cdot|x_{<t}})^2]}$ represent the mean and standard deviation of log probabilities over the vocabulary, respectively. The final score aggregates the $k\%$ lowest-scoring tokens to obtain a robust sentence-level membership score.

3.3 Proposed Method

While Min-K%++ provides a strong baseline, our analysis reveals that it treats all tokens within the selected $k\%$ equally, potentially missing important distributional patterns that could enhance membership detection. We propose a residual score decomposition approach that analyzes local patterns in the normalized scores and applies adaptive importance weighting.

3.3.1 Core Methodology

Our method enhances Min-K%++ through three components: (1) residual decomposition via exponential moving averages identifying tokens deviating from local trends, (2) position-based importance weighting recognizing varying token informativeness, and (3) multi-scale deviation analysis capturing patterns across temporal scales. These combine for nuanced aggregation leveraging local and global distributional characteristics.

Exponential Moving Average Trend Analysis. We decompose Min-K%++ scores into trend and residual components using exponential moving averages (EMA) to identify tokens deviating from local patterns, addressing averaging limitations that obscure informative outliers:

$$\text{EMA}_t = \alpha \cdot s_t + (1-\alpha)\cdot \text{EMA}_{t-1}, \tag{4}$$

$$r_t = s_t - \text{EMA}_t \tag{5}$$

where s_t is the Min-K%++ score at position t, α is the smoothing factor, and r_t is the residual identifying tokens deviating from local trends.